

Colon & Colorectal Cancer — Biomarker Testing Intelligence

Alert Window: Last 7 Days · Generated: 2026-04-12 10:36:52

Situation Summary

Stage Iv: 0 of 6

Biomarker Count: 6 entries

Top Targets: RESEARCH (6)

Breakthrough: 0 entries scored ≥ 4

Research & Guidelines Timeline

Biomarker testing entries ranked by significance score. Higher scores indicate greater clinical relevance. Always discuss with your oncologist.

MEDIUM

All Stages

RESEARCH

VERIFIED

RESEARCH

PUBLISHED APR 09, 2026

⚡ 0

Integration of deep learning and radiomic features from multiplex immunohistochemistry images for reproducible Multi-Outcome prediction in a Multi-Center study of colorectal cancer

This study evaluates the integration of deep learning and radiomic features from multiplex immunohistochemistry images for reproducible multi-outcome prediction in a multi-center study of colorectal cancer.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

RESEARCH

VERIFIED

RESEARCH

PUBLISHED APR 08, 2026

⚡ 0

An algorithm-enhanced stool DNA system improves the differential diagnosis of colorectal cancer versus Crohn's disease in high-risk symptomatic patients

This study evaluates an algorithm-enhanced stool DNA system that improves the differential diagnosis of colorectal cancer versus Crohn's disease in high-risk symptomatic patients.

pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov, pubmed.ncbi.nlm.nih.gov

Appendix A — Patient-Friendly Summary

Plain-language overview for patients, caregivers, and family members

What's happening

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. The results determine which FDA-approved treatments may be available to you. The entries below represent the most significant biomarker testing research and guidelines in this report window.

GENOMIC PROFILING

The Evolving Role for Repeat Molecular Testing in Metastatic Colorectal Cancer

This study evaluates the evolving role for repeat molecular testing in metastatic colorectal cancer.

GUIDELINE

Circulating Tumor DNA as a Biomarker for Recurrence After Curative Resection of Colorectal Cancer Liver Metastases: A Systematic Review and Meta-Analysis

This study evaluates circulating tumor DNA as a biomarker for recurrence after curative resection of colorectal cancer liver metastases through a systematic review and meta-analysis.

LIQUID BIOPSY

A new paradigm in postoperative colorectal cancer surveillance: integrating advanced imaging and multi-omics

This study evaluates a new paradigm in postoperative colorectal cancer surveillance by integrating advanced imaging and multi-omics.

RESEARCH

ROME trial wasn't built in a day: the decade-long journey to randomized evidence in precision oncology

This study evaluates the decade-long journey to randomized evidence in precision oncology as part of the ROME trial.

EARLY DETECTION

From mono- to multi-cellular in vitro models: reconstructing colorectal cancer complexity for translational and personalized applications

This study evaluates the transition from mono- to multi-cellular in vitro models for reconstructing colorectal cancer complexity for translational and personalized applications.

What this means for you

If you have been diagnosed with metastatic colorectal cancer, ask your oncologist: *"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"* Testing should happen at diagnosis — before treatment begins — because results take 1–2 weeks and can shape your first treatment decision. Ask specifically about KRAS, NRAS, BRAF V600E, MSI/MMR status, HER2, NTRK, and TMB.

Generated from structured biomarker research data · 2026-04-12 10:36:52

This is not medical advice. Always speak with your oncologist before making any treatment decisions.

Appendix B — Colon & Colorectal Cancer: Biomarker Testing Reference

What Is Biomarker Testing?

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. A single next-generation sequencing (NGS) panel can check many gene mutations at once. Results typically take 1–2 weeks and should be obtained at diagnosis, before treatment begins.

Key Biomarkers — Test All mCRC at Diagnosis

RAS (KRAS/NRAS) · BRAF V600E · MSI/MMR status · HER2 · NTRK fusion · TMB

Testing Methods

NGS (next-generation sequencing) — checks many mutations at once from tumor tissue

Liquid biopsy / ctDNA — blood test that detects tumor DNA; useful when tissue is unavailable

IHC (immunohistochemistry) — used to test MMR status and HER2

PCR — used for specific mutation detection (e.g. MSI, KRAS)

Questions to Ask Your Oncologist

"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"

"Was my sample taken from the primary tumor or a metastatic site — and does it need to be retested?"

"How long will results take, and will they affect my first treatment decision?"

"Is a liquid biopsy appropriate for my situation?"

Abbreviations

BRAF V600E — a specific gene mutation found in ~8–10% of colorectal cancers

ctDNA — circulating tumor DNA; tumor DNA fragments found in the bloodstream

dMMR — mismatch repair deficient; indicates the tumor may respond to immunotherapy

HER2 — a protein that drives growth in ~2–3% of colorectal cancers

IHC — immunohistochemistry; a lab staining method used to detect proteins in tissue

KRAS G12C — a specific gene mutation found in ~3–4% of colorectal cancers

mCRC — metastatic colorectal cancer (Stage IV)

MSI-H — microsatellite instability-high; indicates the tumor may respond to immunotherapy

NGS — next-generation sequencing; a method that checks many gene mutations at once

NTRK — a rare gene fusion found in <1% of colorectal cancers

RAS — a gene family (KRAS/NRAS) whose mutation status affects treatment selection

TMB — tumor mutational burden; a measure of how many mutations are in a tumor

Key Resources

ClinicalTrials.gov · NCCN Guidelines (nccn.org) · Colorectal Cancer Alliance (ccalliance.org) · Fight Colorectal Cancer (fightcrc.org) · ACS (cancer.org)

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About This Report

This report is generated automatically from publicly available medical databases including PubMed, ASCO Journal of Clinical Oncology, Annals of Oncology, and NCI Cancer.gov. Entries are classified using deterministic rule-based logic and scored by clinical significance. AI-generated summaries are grounded exclusively in the source data provided.

Limitations

This report does not represent a complete picture of all published research. Database coverage, feed availability, and relevance filtering affect what appears. Research findings may not yet reflect current clinical guidelines. Always verify information at the original source before making clinical decisions.

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https://petroai1492.github.io/delta/timelines/coloncancer_biomarker_timeline.pdf